

Harnessing Nonlinear and Optomechanical Phenomena in Silicon Nanostructures

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Abstract Subwavelength silicon nanostructures enable precise control of light and optomechanical interactions in integrated devices. We review our recent advances in mid-infrared silicon photonics and the optimization of Brillouin interactions in non-suspended optomechanical cavities. ©2026 The Author(s)

Introduction

Silicon's broad transparency window (1.1–8 μm) and the suppression of two-photon absorption beyond 2 μm make it a promising platform for mid-infrared supercontinuum generation [1]. However, standard silicon-on-insulator (SOI) technology is constrained at wavelengths above $\sim 3.5 \mu\text{m}$ due to absorption in the silica cladding. Suspended silicon waveguides address this limitation, enabling operation across the full transparency range of silicon [2–4].

In parallel, Brillouin scattering—the nonlinear coupling between optical and mechanical modes—has attracted significant interest for applications in communications, sensing, and quantum technologies [5]. Achieving efficient Brillouin interactions requires simultaneous confinement of photons and phonons, which remains challenging in conventional SOI waveguides due to phonon leakage into the silica substrate.

In this invited presentation, we will report our recent results on mid-infrared supercontinuum generation enabled by engineered silicon nanostructures, as well as on near-infrared optomechanical devices designed to enhance and control Brillouin interactions. These results illustrate the versatility of subwavelength nanostructures for tailoring both optical and optomechanical functionalities in integrated platforms.

Supercontinuum generation in the mid-IR

Supercontinuum generation (SCG) in silicon waveguides is a nonlinear process arising from the propagation of high-power, ultrashort pulses

in a dispersive medium. In the regime considered here, femtosecond pulses are injected in the anomalous group velocity dispersion (GVD) region of the waveguide, where the interplay between dispersion and Kerr-induced self-phase modulation (SPM) leads to the formation and evolution of optical solitons. While SCG can also occur in the normal dispersion regime, the soliton-driven dynamics in the anomalous regime are particularly effective for achieving broad and coherent spectral expansion.

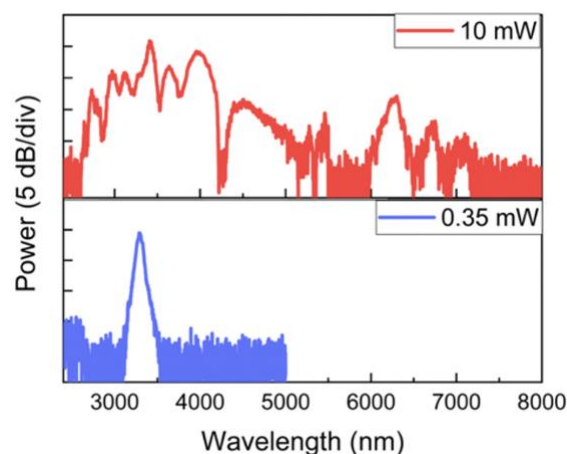


Fig. 1: Experimental characterization of supercontinuum generation in suspended silicon waveguide.

Silicon is an excellent platform for mid-infrared SCG due to its outstanding nonlinear properties. In particular, its Kerr nonlinear coefficient is approximately two orders of magnitude higher than that of silica [6], enabling strong nonlinear interaction over short propagation lengths.

Moreover, two-photon absorption becomes negligible for wavelengths above $2\ \mu\text{m}$, allowing efficient spectral broadening in the mid-infrared. The high refractive index contrast between silicon ($n \approx 3.48$) and typical claddings such as air ($n = 1$) ensures tight optical confinement, reducing the effective mode area and providing significant flexibility for dispersion engineering.

However, in conventional silicon-on-insulator (SOI) platforms, the achievable SCG bandwidth is fundamentally limited by the absorption of the silica cladding at wavelengths beyond $\sim 3.5\ \mu\text{m}$ [2]. To overcome this restriction, alternative platforms have been explored. Silicon-on-sapphire (SOS) waveguides leverage the broader transparency window of sapphire (up to $\sim 5\ \mu\text{m}$), enabling SCG spanning from 2 to $6\ \mu\text{m}$ [1]. More recently, suspended silicon waveguides have emerged as a powerful solution, eliminating the lossy silica cladding altogether. Using this approach, broadband spectra extending from $2\ \mu\text{m}$ up to $7.7\ \mu\text{m}$ wavelength [7].

Subwavelength-engineered or metamaterial silicon waveguides introduce additional degrees of freedom for tailoring the dispersion. Using this approach, we have demonstrated wideband supercontinuum generation, expanding between $2.5\ \mu\text{m}$ and $7.5\ \mu\text{m}$ wavelength (see Fig. 1). These results highlight the potential of

These include the use of bound states in the continuum [8] and geometrical softening [9], which provide enhanced optomechanical coupling with moderate acoustic confinement.

Subwavelength nanostructures offer a versatile route to engineer phonon confinement without removing the substrate. The approach considered here, depicted in Fig. 2, relies on a periodic array of silicon segments defined by their width, thickness, pitch, and filling factor. The key mechanism underlying acoustic confinement in these structures is the control of phonon radiation [10]. A single silicon segment exhibits a radiation pattern analogous to that of an antenna, with quasi-omnidirectional emission in-plane and a structured angular distribution out-of-plane. When multiple segments are coupled, their relative phase of motion plays a critical role. Indeed, out-of-phase motion leads to destructive interference and strong suppression of acoustic leakage. Building on this concept, nanobeam cavities can be designed to enhance optomechanical interactions through simultaneous confinement of light and sound. By operating near the Bragg condition at telecom wavelengths, the optical mode can be localized using a gradual modulation of the waveguide width, forming a high-quality-factor cavity. These structures support a small number of optical

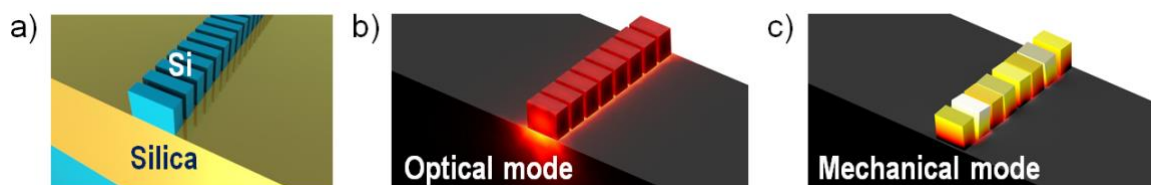


Fig. 2: (a) Schematic view of the proposed optomechanical silicon waveguide exploiting subwavelength periodic structuration. Calculated (b) optical and (c) mechanical modes of the subwavelength silicon waveguide.

dispersion-engineered silicon nanostructures for achieving highly controlled and extended supercontinuum generation in the mid-infrared.

Non-suspended optomechanics

Achieving strong optomechanical interactions in SOI critically depends on the simultaneous confinement of optical and acoustic modes. While optical confinement is readily achieved, acoustic confinement remains a major challenge in non-suspended structures due to phonon leakage into the silica substrate. A widely adopted approach to mitigate this issue consists in partially or fully removing the buried oxide layer, enabling strong Brillouin interactions in suspended membranes. However, suspended geometries increase fabrication complexity, hinder large-scale integration, and reduce thermal dissipation.

Alternative strategies have therefore been explored to realize efficient optomechanical confinement in non-suspended SOI devices.

modes and a rich set of mechanical resonances, all well confined within the silicon despite the absence of a full mechanical bandgap. Simulated acoustic quality factors exceed 10^4 , confirming the effectiveness of radiation-based confinement. Such cavities have been fabricated using standard silicon photonics processes, including electron-beam lithography and dry etching, on SOI wafers with a $500\ \text{nm}$ silicon layer. The optomechanical response is probed via dynamical back-action measurements, showing clear signatures of the optomechanical spring effect as the laser detuning is varied. From these measurements, a vacuum optomechanical coupling rate on the order of $g_0/2\pi \approx 100\ \text{kHz}$ is extracted, in good agreement with simulations. Although fabrication imperfections affect the exact modal distributions and coupling strengths, the experimental results validate the proposed approach and demonstrate that subwavelength-engineered non-suspended structures can

achieve efficient acoustic confinement and strong optomechanical interaction. This provides a scalable and CMOS-compatible pathway toward integrated optomechanical devices without the drawbacks of suspended architectures.

Conclusions

In conclusion, subwavelength engineering of silicon waveguides provides a powerful and versatile framework to control both nonlinear optical and optomechanical phenomena in integrated platforms. In the mid-infrared, dispersion-engineered silicon structures enable efficient supercontinuum generation by harnessing soliton dynamics and dispersive wave emission, while overcoming material limitations through platforms such as suspended and metamaterial waveguides. These approaches allow spectral broadening across a large fraction of the silicon transparency window. In parallel, the same subwavelength design concepts can be exploited to tailor mechanical properties and achieve strong optomechanical interactions in non-suspended SOI devices. By controlling phonon radiation and interference, efficient acoustic confinement can be realized without the need for fully suspended geometries, enabling scalable and thermally robust implementations.

Together, these results highlight the potential of subwavelength silicon nanostructures to unify nonlinear photonics and optomechanics within a single platform. This opens new opportunities for compact and multifunctional devices targeting applications in spectroscopy, communications, and emerging quantum technologies.

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